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Coupling 5G and Wi-Fi 6 Connectivity

Abstract:

The future rests in an interconnected world enabled by high-speed connectivity that allows for applications to function in real-time and at split seconds. Connectivity in the form of cellular and WiFi are seeing constant advancements with the introduction of 5G and WiFi 6. The whitepaper explores the range of opportunities when these two technologies are both complimentary & competitive. The paper extensively highlights each's advantages, drawbacks and the future possibilities/applications that can ensue by leveraging the strengths of both these technologies.

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Introduction to 5G

5G is the next generation wireless communication technology that has evolved from the 4G cellular technology. 5G technology is embracing the need for a smarter network that connects machines, devices, automobiles, safety, and city infrastructures. The highlight of 5G is its ability to accommodate millions of users simultaneously and at a lightning speed of 10Gbps.

It has been proven to serve 100 times the speed of the earlier 4G technology, reduces latency by 10 times, has 1000 times more network capacity catering to efficient IOT connectivity and a more consistent performance across users. Also, it is acknowledged to be ultra-reliable, immensely secure, maintains low energy consumption and has an ability to handle extreme traffic by enabling small cells technology. With these remarkable attributes, 5G will cater to 3 main domains – mission critical control, enhanced broadband & massive Internet of Things and conceive a complete wireless medium of seamless communication.

5G will be built upon the existing 4G advanced technology and will etch a mark in a wide range of industries such as –automobile, healthcare, business, agriculture & many others.

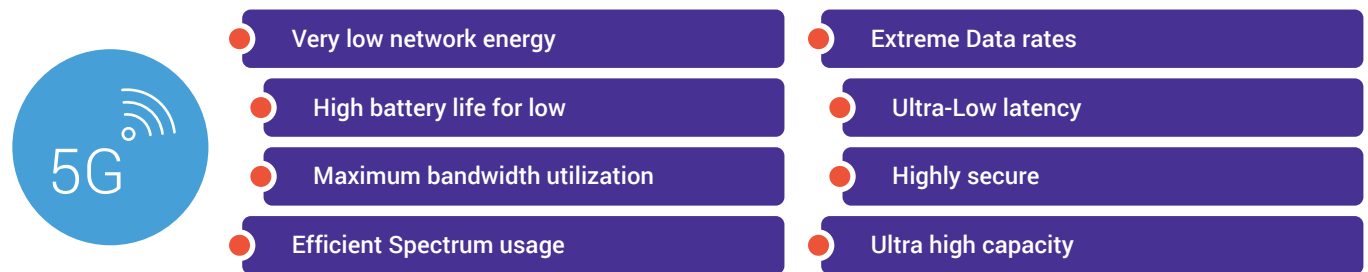


Figure 1



Introduction to Wi-Fi6

A Wi-Fi network is an internet connection that is shared with multiple devices in a home or business via a wireless router by emitting WI-FI (radio) signals. The router with an inbuilt modem acts as a hub to broadcast the internet signals to all the connected Wi-Fi enabled devices. It mainly operates in the unlicensed spectrum.

Wi-Fi 6, also known as 802.11ax, is the next gen Wi-Fi exhibiting speeds almost double the current Wi-Fi standard (Wi-Fi 5) and has a multi-fold capacity. This scales the number of Wi-Fi enabled devices supported, such as smart phones, TV, tablets, IoT systems and computers catering to the increasing demands of density and provide extreme cellular like capabilities to the end user. Wi-Fi is mainly used in relatively small indoor spaces, however with the latest Wi-Fi 6 upgrade, connectivity is promptly expanded outdoors as well.

Over the years, the Wi-Fi technology has evolved, and Wi-Fi 6 is able to yield speeds up to 10gbps. With the recent trend in an increased demand of connected devices, Wi-Fi 6 can split the speed across a whole network of devices attached to the router uniformly. That means more potential speed for each connected device.

Does this ring a bell?! Yes, the Wi-Fi6 Technology can be easily comparable to the 5G Wireless. This whitepaper will emphasize the range of facilities offered by the 5G-Wi-Fi 6 collaboration in modern connectivity.



Evolution of 5G



The internet connectivity has always been distributed to homes via a wired fixed connection with fiber cables. The technology, however, lacks the speeds needed for the modern usage of internet.

The gradual need for speed and high quality of service diminishes the current fiber cabling option and demands a wireless mechanism promising good bandwidth and stability.

The wireless communication has evolved from 1G(AMPS) to 5G over a span of 3 decades. The transformation has been massive from analog voice to Era of Digital data in 2nd generation (GSM GPRS). Thereon, the focus has been on speed, quality & simultaneous video/audio. From 3rd (UMTS) to 4th generation (LTE) these attributes have been getting stronger and promise excellent streaming experience.

The 5th generation wireless technology will be the ultimate game changer wherein all the real-world applications and devices in different domains will be connected with super high-speed internet, lowest possible latency and unlimited data capacity.



Evolution of Wi-Fi 6

In 1999, the Wi-Fi technology was launched to support a maximum speed of 11mbps. Over the years it has evolved extensively to support faster speeds and accommodate more devices to increase the connectivity.

The biggest development in the Wi-Fi technology materialized when the MIMO (Multiple Input Multiple Output) in 4th Generation was replaced by DL MU-MIMO (Downlink Multi User Multiple Input Multiple Output) in the 5th Generation and the most recent Wi-Fi 6 introduced the DL/UL MU-MIMO (Downlink & Uplink Multi User Multiple Input Multiple Output). As the name suggests, the evolution in terms of the number of users availing the services with uniform quality throughout the streaming and speeds has elevated multifold.

The increasing demand for ultra-high-speed games, ace quality video streaming and immense bandwidth demanding applications, makes the necessity for faster internet speeds a constant requirement.

Apart from speeds, the number of devices which the router can accommodate is the next path to evolution. The usage of OFDMA system (Orthogonal Frequency Division Multiple access) along with MU-MIMO wireless technology uplifts this in the latest Wi-Fi 6 routers catering to the modern consumptions. The 5th generation onwards, OFDMA technology allows users to be allocated in both time and frequency access enabling higher number of users to access the internet at the same time with homogeneous quality. The evolution from Wi-Fi 5 to Wi-Fi 6 is not the drastic change in speeds but is the ability to accommodate more and more devices with highly reliable speeds and enterprises to bare high-density environments.

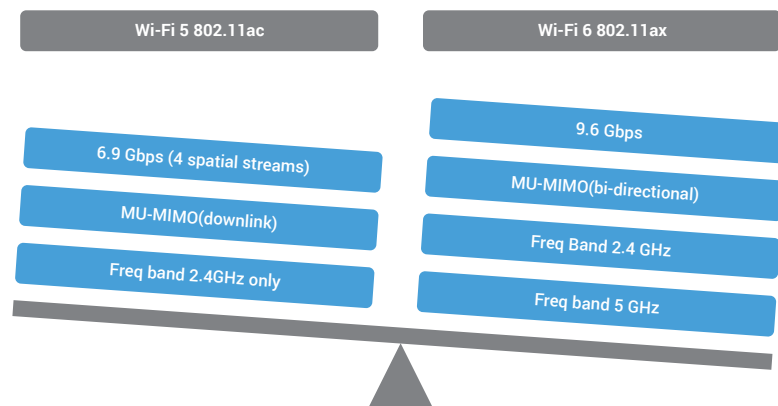


Figure 3



Detailed explanation on 5G and Wi-Fi6 (Features)

5G

5G promises to deliver greater throughputs, ultra-low latency, ultra-high reliability, consistent & uninterrupted connectivity, and higher mobility range with utmost privacy. It is designed to use the licensed band of the frequency spectrum and can accommodate millions of devices simultaneously through carrier aggregation, powered by the innovative Advanced RAN Coordination which allows full deployment flexibility and uncompromised user throughput.

It operates in –

Low bands (1G Hz) for higher range feature like mobile broadband and massive IOT.

Mid bands (1-6 GHz) for eMBB and mission critical services enabling wider bandwidths.

Higher bands (>6GHz) mmwave band for its critical performance, better coverage, and is compatible with LTE, WI-FI and other technologies.

The vision of implementing autonomous vehicles requires data to be transmitted with ultra-low latency and a real time turn-around time in order to avoid mishaps. Theoretically, this will stop a vehicle much quicker than a human operated vehicle, as the reaction time of the vehicle will be 1000 times faster than the average human. Connected cars technology demands high bandwidth and ability for multiple vehicles to be connected to each other. Both these key features are possessed well by 5G.

Not only does 5G support vehicle-to-vehicle communication (V2V) but it facilitates vehicle-to-everything (V2X) as well. To meet the higher bandwidth and lightening speeds, 5G deploys transmitters across different infrastructures such as traffic lights, lamp posts, buildings, etc. enabling the revolution of the smart city technology.

It can provide the speeds and necessary bandwidth for internet connections in local and personal networks, such as small office spaces. 5G-enabled Customer Premises Equipment (CPE), will allow businesses to operate in areas where no fixed high-speed internet is available, such as rural areas and in older buildings.

The mission critical mobile communications ensure resilient connectivity and extremely reliable data paths for high security government operations, first responders, military, and other crucial industries.

5G possesses Cognitive Radio (CR) - a form of wireless communication in which a transceiver can intelligently detect which communication channels are in use and which are not. It instantly moves into vacant channels while avoiding occupied ones. It does not cause any interference to the licensed user.



Wi-Fi6

The Wi-Fi 6th generation allows routers to communicate with numerous devices at once, lets routers send data to multiple devices in the same broadcast, and allows Wi-Fi devices schedule check-ins with the router. Together, these features should keep connections strong even as more and more devices start demanding data.

The biggest strength of Wi-Fi6 is not the change in speed but the ability to handle more and more devices attached. Not just the capacity but also ensure every device is strongly connected no matter how many new devices attach to the router. In today's Wi-Fi 5, when the number of devices attached increases, the traffic load increases, and the internet speed drops drastically giving a bad user experience.

The main technologies enabling the router to overcome the above issues are

- MU-MIMO Multiuser Multi Input Multi Output
- OFDMA – Orthogonal frequency division Multiple access

802.11ax creates MU-MIMO connections so that with downlink MU-MIMO an access point may transmit concurrently to multiple receivers and with uplink MU-MIMO an endpoint may simultaneously receive from multiple transmitters. 802.11ax supports up to eight MU-MIMO transmissions at a time, up from four with Wi-Fi5.

Up to 8 devices can have separate channels and each can carry data with high speeds and quality explaining how Wi-Fi6 routers can communicate to all attached devices simultaneously than broadcasting to each device one by one.



Figure 4



The main difference between an OFDM (used in 4th and 5th gen Wi-Fi) and an OFDMA (Orthogonal Frequency Division Multiple Access) system (Wi-Fi 6) is that in the OFDM technology the users are allocated on the time domain only while using an OFDMA system the user would be allocated by both time and frequency. It allows one transmission to deliver data to multiple devices simultaneously.

Example: Consider the MIMO as an Uber ride. The ride is exclusive for the passenger to reach their destination in the best speed and route. Now, with Uber share ride (OFDMA), the App(router) can calculate if a ride is half empty and can accommodate another passenger on the way without compromising speed and time and saving energy.

Wi-Fi 6 operates in the current standard frequencies (2.4 GHz and 5GHz) and future bands (1GHz and 6GHz) when available promising a massive new allocation of unlicensed radio spectrum offering bigger channels, stronger signal & faster data. With more of the spectrum available, Wi-Fi 6 can divide the bandwidth into narrower

sub-channels, providing more avenues and access points creating a small sub-network on its own.

Wi-Fi 6E

A Wi-Fi router is handling massive amount of cellular data every minute but is allocated only about ~600MHz of unlicensed spectrum. The congestion has been increasing for some time, but when the pandemic hit, the amount of data on Wi-Fi networks escalated, making the performance worse. The case for increasing wireless bandwidth and capacity has never been greater.

This in-turn took a huge toll on the efficiency and the ability for the devices to stay connected especially because the bandwidth has remained constant while the number of devices attaching to this bandwidth has increases exponentially.

The Wireless Internet Service provider's association along with all the big tech companies are working towards utilizing the 6GHz frequency band to regale the customer with significantly better user experience and create paths for future endeavors. This extension

of Wi-Fi 6 will be called Wi-Fi 6E. It has recently been reviewed by FCC and all future certified Wi-Fi devices will have this feature in built. With this upgrade, 5G and Wi-Fi 6 combined devices can work seamlessly indoor and outdoor in the same frequency bands.

The Wi-Fi 6E version will be able to cater to almost ~70 clients and guarantee a uniform speed of up to 2 Gbps per device! It will further reduce interference and provide improved responsiveness.

Wi-Fi 6 and Wi-Fi 6E devices will be inter-operable making it a low-cost affair and easier to reach common man without much of an expense.



Architecture - 5G and Wi-Fi 6 coupled

Creating a world with the best of 5G features and Wi-Fi 6 capabilities will be the way forward creating a whole new level of connectivity experience. 5G is mobile network's fastest and Wi-Fi 6 is Wi-Fi's highest grade. While the former is known for large scale outdoor arenas & low latency, the latter is known for its efficiency in smaller indoor spaces.

In the current age, a user promptly switches from their Cellular Data connection to Wi-Fi while moving indoor and vice versa when stepping out.

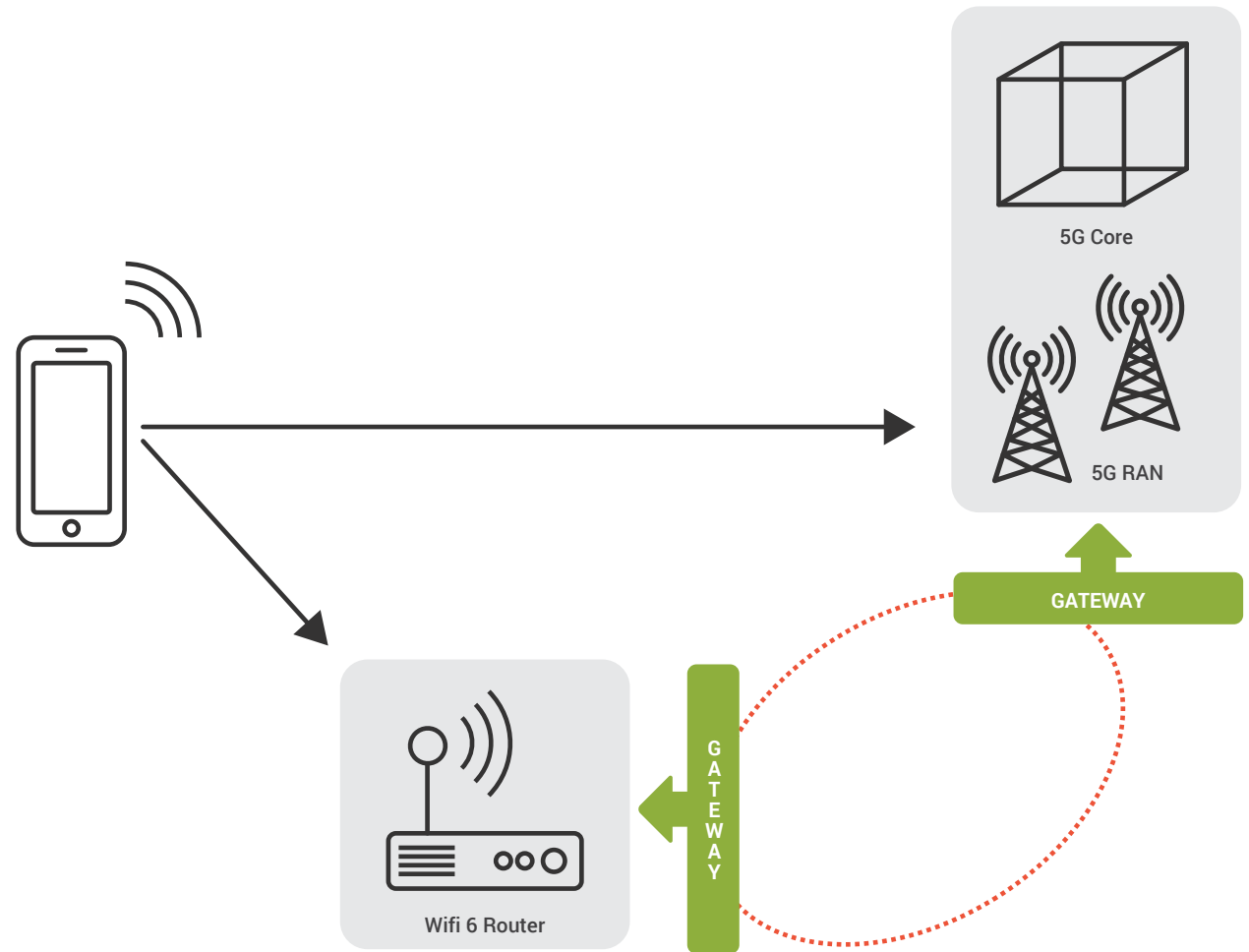


Figure 5



One way of achieving a 5G and Wi-Fi 6 coexistence is by network operators offering a new interface which would enable the aggregation of the Wi-Fi radio signal just as other radio signals and offer the entire mobile with Wi-Fi experience as a single solution. The other option is Wi-Fi routers integrating 5G modems into their devices and allowing connectivity to the internet using the existing wired configuration. Alternatively, network operators can help users switch dynamically between Cellular and Wi-Fi based on congestion, speed, and quality of service desired. This will allow mobile network operators gain access to Wi-Fi networks and its consumers to provide a better service.

The 5G and Wi-Fi 6 collaboration could involve one instrument - a router or a smart phone and work seamlessly to provide connectivity irrespective of the place & activity, offering the benefits of them both. The ideology is for the user to subscribe to services based on the need. The network assigns Wi-Fi 6 or 5G attributes based on the network's matrix, user's current task and requirements, ensuring the best experience.

3GPP has announced the acquisition of unlicensed spectrum of 6 GHz band, which is the same as the Wi-Fi 6E spectrum, marking a grand entrance for the 5G-Wi-Fi 6 co-existence. This development enables access to broadband that wasn't available in previous generations, resulting in faster processing and stronger connections.

In the end, Wi-Fi enterprises will be able to have better visibility on mobile networks and operator provided 5G subscriptions. This integrated solution will have an enormous impact on the end user by plying between Wi-Fi and Cellular Network signals delivering the best of both worlds.



How they complement each other

The 2 main technologies for connectivity for customers across domains are Cellular and Wi-Fi. Because Wi-Fi has a lower cost to deploy, maintain, and scale—especially where access points need to serve more users—it will continue to be the predominant technology for home and business environments. This provides great support for dozens of data-hungry devices, like PCs, tablets, smartphones, streaming devices, TV sets, and printers, which must all connect to the network. Thanks to its longer range, 5G will be used for mobile connections, like smartphones. It will also be used for connected cars, smart city deployments, and even for large manufacturing operations. When these solutions are integrated together, their combined abilities can bring out the best in modern connectivity. Wi-Fi in general uses the unlicensed spectrum which implies low operational and setup cost.

On the contrary, in 5G, the usage of licensed spectrum establishes secure and reliable connections. However, the main reason that 5G will not oust Wi-Fi

connectivity, both at home and in the public sphere, is the connectivity restrictions presented by chips in our internet-enabled devices. All current consumer devices rely solely on a Wi-Fi or 4G LTE connection. Indeed, many tablets, in-home IoT technology, and smart TVs to name a few, are exclusively restricted to the more reliable spectrums that Wi-Fi offers. They can work together in managing the spectrum efficiently based on the need of the internet usage.

The next iteration of Wi-Fi will support mesh networking. Mesh networking, in simple terms, is a network infrastructure made up of a series of interconnected nodes; all connected directly and dynamically to as many other nodes in the network as possible. To create the mesh networking, there are several Wi-Fi access points identified. This is very similar to how the 5G millimeter wave will be deployed. It's a big opportunity to drive millimeter wave indoors as a private network. This is another scope where 5G and Wi-Fi 6 can interwork to cater to best user experiences.

With respect to configuration, a well configured SIM card with 5G capabilities can be easily installed and one can subscribe to its exceptional services. This could make tasks such as comms configurations easier. 5G is all about connecting all types to devices, but devices are designed to not support the insertion of SIM cards or ESIM cards and need to be connected through Wi-Fi. Wi-Fi-6, with better network speed quality than before, can be integrated well with the 5G devices and address new market demands for applications with extensive throughput and QoS requirements.

Wi-Fi 6 delivers 40% higher throughput, lower latency and simply connects more users and more devices, offering a high standard for anyone working, studying, or relaxing from home. In addition, Wi-Fi 6 uses many of the same technologies as 5G, making it ideal for 5G mobile data offload. With this collaboration, devices can either connect directly to the 5G network or via a local WLAN using Wi-Fi access points and existing wired infrastructure. 5G brings speed and mobility on the



table, while Wi-Fi 6 can provide ubiquitous connectivity and traditional local area network access technology.

In-built Wi-Fi inside laptops is a future possibility, enabling laptops with 5G and Wi-Fi 6 combined routers that efficiently utilize the spectrum. Connected laptops with in-built SIM cards are already in the market targeting users who need uninterrupted connectivity on the move. Now with Wi-Fi inside, consumers can use laptops with best of Wi-Fi and Cellular capabilities.

We expect to see both new technologies leveraged in a complementary way across different industries with their unique strengths.

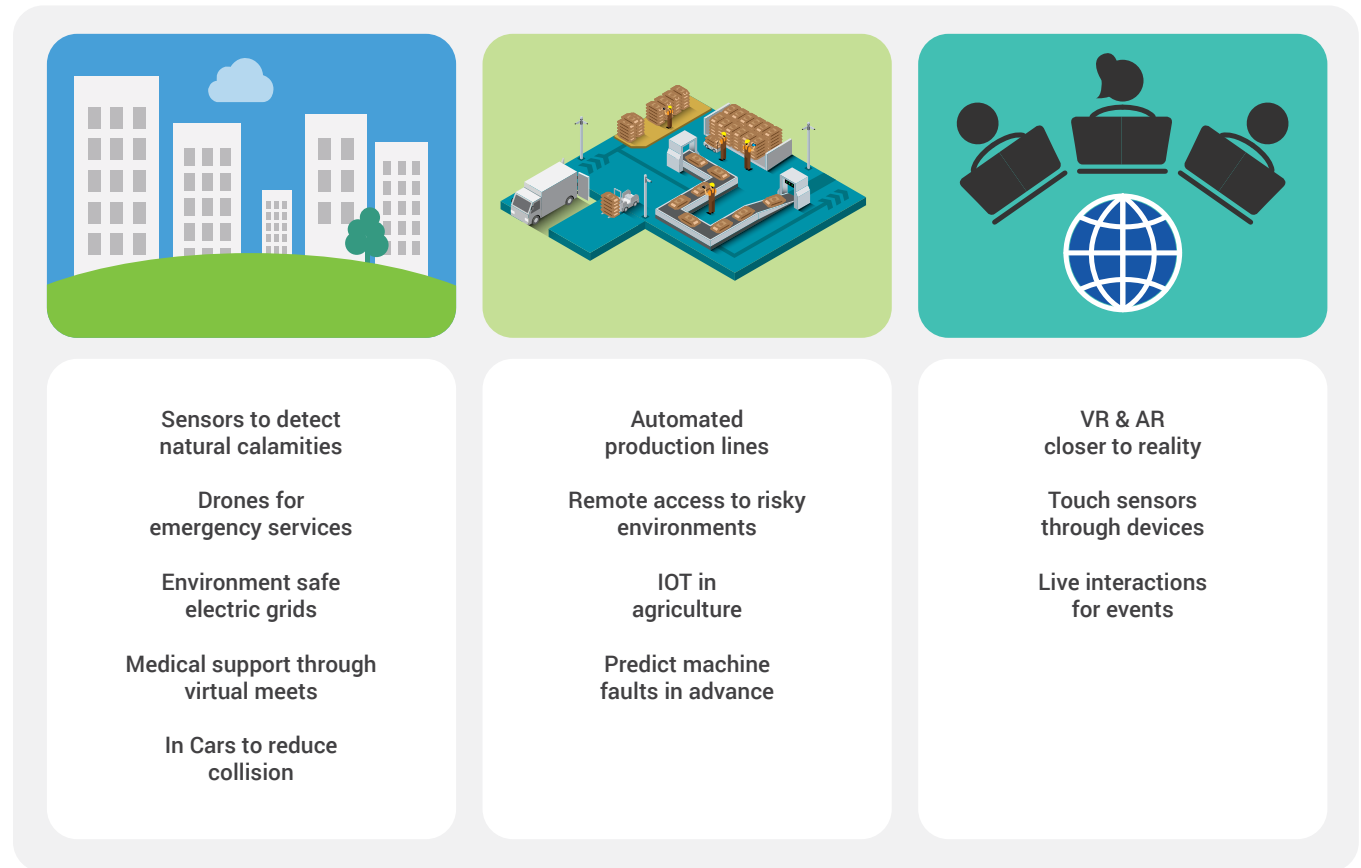


Figure 6



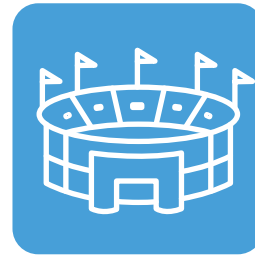
Use cases

The potential use cases of a cognitive 5G -Wi-Fi6 technology can be as imaginative as possible.



Weddings

Weddings are conducted gathering many guests. The indoor venue with a Wi-Fi router is most times unable to handle the traffic from multiple smart devices. The live streaming is often made available to the guests away from the main stage. But this can be converted into a great experience if Wi-Fi 6 routers had 5G integrated and allowed multiple connections streaming live videos onto smart phones simultaneously.



Sports: Security

Around the world, any form of sports is widely watched. 5G wireless technology facilitates billions of cellular devices that can be connected with uniform quality of service. But this heavy traffic and outdoor environment brings a huge risk to the security of information being transmitted. This can be diminished to a large extent by using Wi-Fi 6 enterprises, which by nature has a better control over the consumers. If the traffic from 5G networks can be redirected to smaller internal Wi-Fi 6 groups, the connectivity can be uniformly maintained with minimal threats.



Sports: Live streaming

Sports enthusiasts love watching their favorite game live at a stadium. The live streaming is often played on a big screen telecasting the replays and decisions whenever needed. Imagine if this could be seen on your personal smart phones and tablets, all connected at the same time with well-equipped quality and speeds. The user can rewind and watch their favorite part, check on scores & zoom to replay anytime without relying on the big screens. All this would be possible if connectivity can accommodate such huge traffic and provide the best quality channels to each device attached via the 5G and Wi-Fi 6 integrated devices.





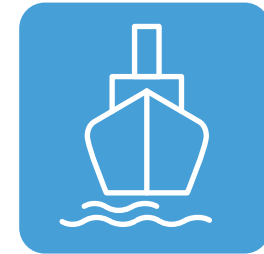
5G to Wi-Fi6 Retail app

The next retail experience allows customers to try on displayed apparel virtually using a seamless integration between mobile data and Wi-Fi 6. The store can provide all the product details to its customers in a secure manner through its personal Wi-Fi system onto the individual's smart phone using the 5G collaboration. Wi-Fi 6 and 5G can coexist and exhibit enhanced mobile broadband through Augmented and Video Reality (AR/VR). During the ongoing pandemic, this could be the best solution for retail experiences that maintains all safety precautions.



Education

The best example of a 5G-Wi-Fi 6 collaboration would be its application in the field of education. Today, most students use their laptops that are connected to the classroom Wi-Fi in schools. Teaching and learning has become an elevated experience with state-of-the-art facility and real time interactions. When these devices can avail outdoor 5G services, as well as indoor Wi-Fi 6 connectivity, students can learn on the move without any disconnection.



Ship to shore communication

The shipping industry has a major concern communicating with ships nearing the coast or parked at a distance due to the long communication range needed. Critical messages often need to be exchanged on highly secure channels between the ship and the shore. Internally, within the yard on the shore, heavy volumes of data need to be transmitted and received between departments. Keeping both volume and range in mind, 5G-Wi-Fi6 collaboration works best. The former for communicating with the ship and latter for internal connectivity at the shore.





Hi-tech Apparels

Soon there will be 5G enabled ultra-sensitive and thin sensors installed in apparels which can monitor health conditions such as blood pressure, O₂ levels, heart rate and other parameters. These sensors can connect to Wi-Fi 6 networks and in turn to the IoT personal sub-networks created in the personnel space providing a combined data of lifestyle information. The applications of these inbuilt sensors can be widened to include tracking persons in remote and dangerous environments such as submarines, mines, under water, etc.



How they compete with each other

Every upgrade in the 5G architecture would need an equivalent upgrade in the device and its hardware implementation. For example, today a smart phone supporting 5G services may possess 5G modem, can operate in lower bands, work in mm wavelengths, duplex in time division but can work only in the NSA (non-standalone mode). The next upgrade in 5G devices may be able to handle freq division duplexing and/or operate in SA (standalone) mode. Hence, for every major change in smart phone 5G capabilities, new devices will have to be purchased. Similarly, for all Wi-Fi 6 devices too.

Which one will oust the other – Wi-Fi 6 or 5G? The reality is that networks are now about connecting more and more endpoints, that's where the battle lines will be drawn between the two types of networks.

When Wi-Fi 6 came up with the mesh network concept, it deployed multiple access points to create a personal sub network to handle the growing number of devices attached to it. This turns the network into a series of wirelessly connected routers, which in turn offers greater speeds and reliability for relatively low costs. Now, 5G

technology researchers are implementing the mmwave similarly to create a comparable experience. But the cost involved is much more because 5G devices are highly complex, expensive, and not backward compatible.

With the number of IoT devices forecast to increase rapidly over the next few years, businesses will need to consider how to link all devices to existing networks – through easy & cheaper Wi-Fi 6 or secure & high speed 5G.

The varying market requirements and the constant need in enhancement of the upload and download capabilities makes it feasible for the intermixing of these two technologies using their strengths or mask their shortcomings.

Cost

5G is known for its best coverage, range, and speeds. These traits are achieved by deploying specially designed network components, intricate antennas and complex software algorithms which require an enormous investment. Not only do network operators need to

install these components but also the consumers need to possess laptops/PCs/smart phones which support this technology. Every 5G subscription will have a price tag as well, making this a high-cost facility. 5G networks needs a lot of base stations to use higher frequency millimeter waves eventually leading to environmental degrade. It means that 5G's ability to penetrate walls and receive signals indoors is worse. Operators face a high cost in deploying infrastructure and security & privacy mechanisms that are imperative.

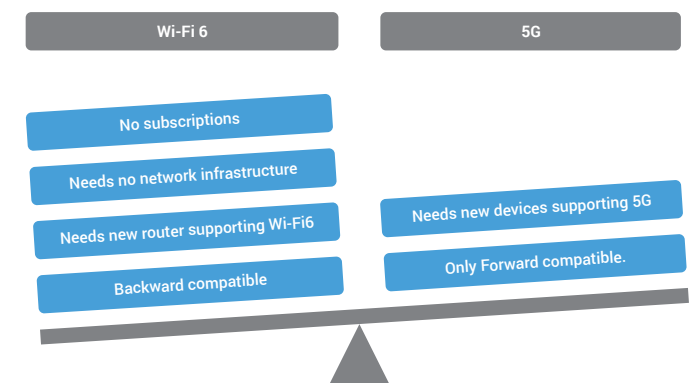


Figure 7a



Wi-Fi 6 on the contrary needs minimum upgradation from Wi-Fi 5 on the router front to provide the enhanced features and speeds. Also, they are backward and forward compatible making it easily available for new and existing customers.

Radio Traffic

Moving forward, networks will face a continued dramatic increase in the number of devices, exponential increase in the total global IP traffic, and a diverse range of new technologies will depend on connecting to Wi-Fi 6 routers. This trend is already visible during the ongoing pandemic. The number of devices connected to their indoor Wi-Fi networks for work/social engagements has resulted in a toll on router capabilities. 5G will be able to cater to millions of devices connected with constant services due to its high bandwidth availability for all users. Conversely increased bandwidth also means less coverage, thus requiring carriers to install more transmitter towers leading to congested radio frequency spectrum. Hence, when the 5G main and Wi-Fi 6 mains are perfectly matched, we can gain a balanced connected environment.

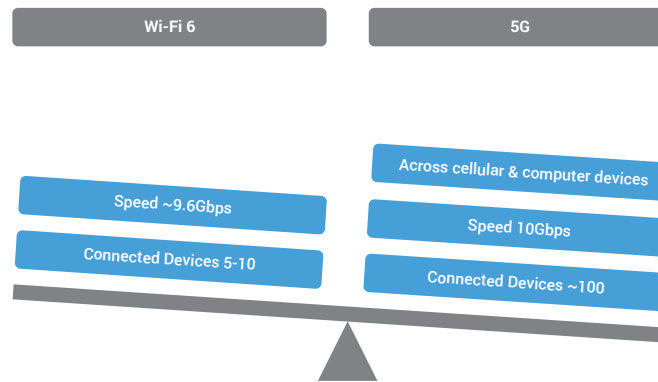


Figure 7b

New network infrastructure demands added privacy and security algorithms.

The connected device and the network operators ought to invest in higher critical security mechanisms to protect their data and privacy. While this is a must for the newly arriving 5G smart devices, experts estimate that the 5G networks do not deploy adequate security measures and is hack prone due to the increase in web traffic and the complexity involved.

Wi-Fi 6 systems lacks the basic security layer as well, as the authentication by the SIM card any cellular networks provides, and its usage of unlicensed spectrum makes it more prone to hacks and interference. Because they communicate with each other over airwaves, your devices and personal information can become vulnerable to hackers, cyber-attacks, and other threats. This is especially true when a user connects to a public Wi-Fi network at places like a coffee shop or airport. However, Wi-Fi 6 has SRT (Spatial Multiplexing Technology) which solves the signal interference caused by cross coverage and improves its anti-interference ability.

Nonetheless, both can offer a flexible deployment architecture that can go from an unauthenticated network all the way up to a very secure enterprise or government network which is evolving every year.



Connected cars and city infrastructure

Perhaps the best modern evolution in connectivity is to implement a smart city where cellular phones and PCs are not the only devices to connect to internet but to be able to connect to every object on the road, buildings, vehicles and create an environment of uniform, seamless, uninterrupted, high speed data connectivity. The major factor driving this forward will be 5G technology. 5G can even couple with artificial intelligence and robotics to bring further revolutions in the next generation wireless connectivity. 5G services will enhance the city in the field of education, healthcare, and energy but the three sectors that are the current major areas of focus in the market are the transportation, public safety & security, and citizen services sectors. While Wi-Fi 6 is widely used in streaming and recording of multiple camera feeds inside and outside of the car to promote driver safety. It is also going to give a tough competition to 5G in this domain owing to lesser cost associated with usage of unlicensed spectrum as well as infrastructure. However, Wi-Fi 6 cannot beat 5G in the range especially in huge parking lots where the Wi-Fi access points are not easily accessible.

The requirement of high bandwidth increases the congestion and impact wireless performance on the roads, in traffic jams, in parking garages and even dealer lots. Wi-Fi 6 battles this issue with OFDMA helping multiple users, with varying bandwidth needs, to access the channel simultaneously & BSS (basic service set) technology which can recognize non-essential traffic to improve bandwidth.

Another upper hand of Wi-Fi 6 in connected cars is the power efficient mechanisms used where the car can go to sleep for a set amount of time and then wake up as needed for Wi-Fi connectivity.

5G has excellent download and upload performance with the help of MU-MIMO technology. However, Wi-Fi 6 is battling to provide better upload speeds.

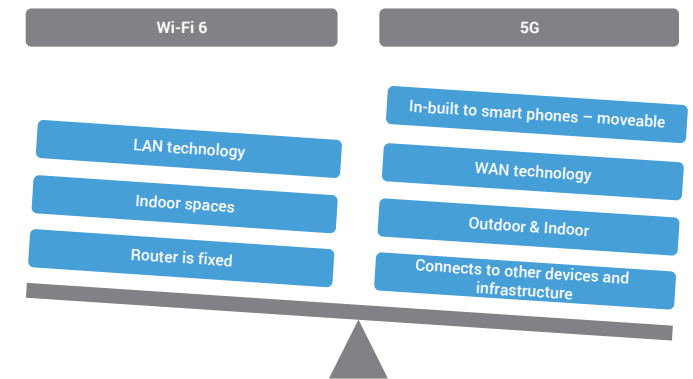


Figure 7c

Power consumptions

Both 5G and Wi-Fi 6 drain a lot of battery due to their continuous dealing with high volumes of data. Wi-Fi 6 has a special feature called Target Wake time. This feature cannot be exploited in every device. For example, smart phones and laptops will need to connect to Wi-Fi all the time for uninterrupted internet access. However, for smaller IOT devices or sensors and fitness gadgets which need not have regular updates, we can tune them with a wake time lowering the battery usage.



Security aspects

Wi-Fi is traditionally more secure than publicly available cellular networks, an important caveat for end users, and those enterprises that conduct business on available connections. With a surge in the number of customers/devices/information out there trying to connect, the connection becomes vulnerable to hackers, especially since it lies in the unlicensed spectrum.

The latest protocol of WPA3 (Wi-Fi Protected Access 3) has a new set of password algorithms. All Wi-Fi 6 routers must include this protocol to clear their certification which makes remarkable effort towards guaranteeing the authentication. This means -

- Individualized encryption (add encryption even when user has chosen a less complicated password)
- Boosts the cryptographic strengths for crucial fields such as defense, medical and any other agency exchanging extra sensitive information.
- Wi-FiEasy connect – process of connecting to a secure wireless network for non-display devices by scanning QR codes.

- Wi-Fi Enhanced Open is the most recent WPA3 protocol. It provides encryption and privacy on open, non-password-protected networks in areas such as cafes, hotels, restaurants, and libraries.

On the other hand, 5G is potentially susceptible to cyberattacks because of its extreme flexibilities and capabilities. Even the core networks are managed by software, thus increasing the visibility to hackers. Future 5G networks will need to support backward compatibility to legacy networks which eventually open the system to invaders that facilitate attacks against devices by attempting to trigger the latest device to connect to older-generation cellular technology that uses weaker security.

Nevertheless, 5G is more secure than any of its predecessors. It encrypts more data, so there is less available to be intercepted. It has 4 new entities to beat the security challenges

- The Security Anchor Function (SEAF)
- Coordinates authentication between UE and Home

network Authentication Server Function (AUSF)

- Authenticates UE using Keys Unified data management (UDM). This entity is responsible for choosing the right Authentication method based on Subscriber Identity and configuration policies.

- Subscription Identifier De-Concealing Function (SIDF)
- decrypts the Subscription Identity. For example, IMSI using a private key, devices which will calibrate a 5G and Wi-Fi 6 integrated solution will imbibe the best of encryption protocols and authentication mechanisms as mentioned above in such a way that the data transfers are equally safe in the licensed and unlicensed spectrum.

The security shortcomings of both the individual technologies will be camouflaged by each other and can fight the cyber-attacks making it a more secure and reliable connectivity for the end users. Therefore, pushing the wireless network industry and the Wi-Fi enterprises to constantly improve their security mechanisms at every upgrade to ensure an affordable and safe environment is of paramount importance.



Cost, infrastructure, and integration

The choice on how 5G with Wi-Fi 6 solutions can be integrated is based on how their internet will be provided - via a wired connection or via a next-generation hybrid 5G router that will emulate traditional fixed-line internet. Whilst it may take time for 5G to become a standard for mainstream devices, a hybrid solution such as a Wi-Fi router with an integrated 5G modem can bring additional flexibility. Alternatively, one can integrate Wi-Fi into the 3GPP control layer, application, or operating system. This combination allows multi-gigabit internet connectivity to be deployed without having to lay a physical cable, thereby greatly reducing deployment cost and time.

However, deploying 5G networks and upgrading smart phones & laptops to support 5G involves a high cost. The Wi-Fi routers with Wi-Fi 6 capabilities are expensive too but are deployed in large scale and do not need much network engineering for every household to have them. They can also be controlled by centrally monitoring users and devices accessing network.

5G wireless networks were designed to cater to a large number of users across vast landscapes using high power and expensive wide area connectivity using macro cell architectures. Using this path of integrated solutions, network operators can reduce this cost and the usage of high-speed cell reselections. Every upgrade in technology, be it in Wi-Fi 6 or 5G calls for an upgrade in the hardware such as antennas, gateways, and base stations. Hence, a device change is inevitable to be able to support the 5G and Wi-Fi 6 integrated competence. Fixed wireless access is the most efficient way to combine 5G and Wi-Fi 6, providing wireless connectivity to homes and businesses through radio links between two fixed points without the need for fiber and cables for last mile connectivity. The expense entailed to install fiber optics and maintain them is immense, especially in rural environments or overly congested surroundings.

The convergence of 5G and Wi-Fi 6 is an ideal solution to populate the high data rate capable wide-area connection to an office, residence, or venue, which solves the challenge of disrupting the city infrastructures setting up fiber cables. Once the building is connected, Wi-Fi is an ideal technology to translate that carrier-specific wide-area connection to a local area connection over the unlicensed spectrum.



Conclusion

The 5G network is suitable for open, mobile, and intensive device connections. Wi-Fi 6 is preferred for home networks and private enterprises. Therefore, 5G and Wi-Fi 6 combined solutions are perfect to be developed on the road of mutual progress, complementation, and integration. From 5G enabled smart phones & smart TVs to the world of IOT & voice assisted Wi-Fi 6, it is surely the best way forward to have them coupled in a balanced and productive way.

With the data traffic expected to grow significantly and an increasing proportion of devices accessing these services, blurring the boundaries of the best connectivity technologies is the most favorable evolution one could imagine.

To summarize, here are some of the key factors which will contribute to making 5G and Wi-Fi 6 convergence a huge success in this era of modern connectivity where every device can connect to each other to form a cloud of internet.

- Availability of the inter-operability to use best of both technologies
- Dedicated high bandwidth
- Unified LAN and WLAN infrastructure bringing more control and security
- Complementing developing and initial setup expenditure
- Efficient usage of spectrum
- Effective power management

Ultimately, the competition between 5G and Wi-Fi 6 technologies offers important benefits by enabling greater flexibility for users to mix-and-match the technologies, business models, and spectrum usage models to best fit their needs.

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